Affect of Psilocybin Mushroom Consumption on Difficult Mental Arithmetic Score

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Introduction: Motivation, Background, & Research Question

When crafting our research question, we began with an interest in whether consuming psychedlic mushrooms would have an effect on ability to perform some sort of mental task. We had prior knowledge that psychedelic mushrooms have a psychological effect often paradoxically associated with both enhanced creativity and distorted cognitive function (Bonnieux et al.)[1]. As such, we were curious whether enhanced creativity could aid in performing arithmetic or if distorted logic would hinder one's ability to do mental calculations. With the variables pre-provided by the Islands software, we choose our explanatory variable as the consumption/lack of consumption of Psilocybin Mushrooms and our response variable as the score participants received on said difficult mental arithmetic exam. Ultimately, we crafted the following research question to guide our study: do the scores on a difficult mental arithmetic exam that adults 18+ on the Islands receive differ between those who consume 10 grams of Psilocybin Mushrooms compared to those who do not consume any Psilocybin Mushrooms?

[1]

[1]Bonnieux, Justin N et al. "Psilocybin's effects on cognition and creativity: A scoping review." Journal of psychopharmacology (Oxford, England) vol. 37,7 (2023): 635-648. doi:10.1177/02698811231179801

Methods

Planned study design: We will be using an experimental study design. Half of our experiment's participants will be randomly assigned to receive 10 grams of Psilocybin Mushrooms, while the other half will be assigned to the control group that will not receive any Psilocybin Mushrooms.

The target population/participants: We will attempt to generalize our results to all adults (defined as over 18 years old for purposes of this study) on the islands.

The inclusion and exclusion criteria: Our only exclusion criterion is individuals under the age of 18, due to the ethical dilemmas that providing psychedelics to individuals under 18–like children–might pose.

The sampling methods: We will be using random sampling and random assignment. We are going to use two random number generators to select a city and a household, and then randomly pick an adult from that household to be a participant. From all the individuals selected, half will be assigned to the treatment group and the other half will be assigned to the control group. We will try to roughly sample the same number of individuals from each island.

The intervention(s): The consumption/lack of consumption of Psilocybin mushrooms (2 levels).

The outcome(s) to be measured: The score received on a difficult mental arithmetic exam (quantitative).

Other demographic variables collected: We will record information on age, gender, wealth, and island of residence for each participant in our study.

The sample size calculation and justification: See "Summary of Power Analysis and Sample Size Determination" section

The randomization (random assignment) methods: We are using random sampling and random assignment, as described above under sampling methods

The blinding methods: We will not be using blinding methods, since us (the researchers) knowing the assignment of the explanatory variable has no effect on the response variable, and we didn't have an option to use a placebo for a single-blind test (and using something else as a placebo could have an effect we are unfamiliar with due to the digital format of this study).

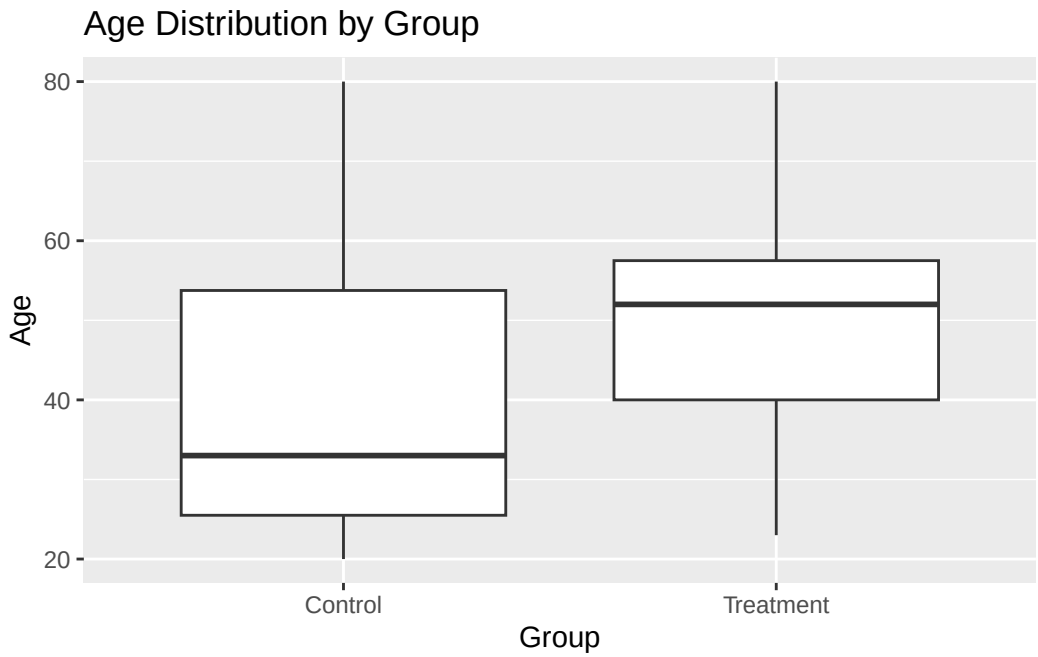
The study implementation: We will randomly sample individuals from the islands as outlined above, half of the individuals will take mushrooms (1 minute), and then after the conclusion of the consumption, those individuals will take the 'Mental Arithmetic Difficult' test (4 mins). The other half of randomly sampled individuals will only take the 'Mental Arithmetic Difficult' test (4 mins).

The statistical methods used to analyze the data: Two-sample T-test.

Summary of Power Analysis and Sample Size Determination

We began by analyzing what sample size would give us a power of at least 70% (a 70% chance of correctly rejecting a false null). We based this off of preliminary collected data where we found a sample mean difference of 6.6 points higher for participants who consumed psilocybin mushrooms than for those who did not. We analyzed three scenarios, and found that to obtain both a significant result when the true population difference in means is 6.6 points and a power of approximately 0.7, our sample size should be around 17 per group. However to account for the natural variability that comes from the randomized data collection process, we ultimately decided to collect 22 observations per group (44 total). Based on this sample size and estimating based on a true alternative difference of 6.6, we estimate the actual power to be 0.807. Since we planned to conduct a two sided hypothesis test, and therefore we were not too worried about the extra directionality that pilot data might suggest when doing a one-side test, in the interest of time we chose to include our pilot study data into the final dataset. However, if this were a real study and not in the islands, we would have most likely collected new random samples as pilot data is usually not included in the full dataset.

Also, as previously mentioned, we used random sampling in order to allow for generalization to the population of individuals on the Islands. We evaluted numerical summaries to reflect the extent to which our generalization worked. We ended up with the exact same ratio of 0.6923077 male to female participants between our treatment and control groups, which were the same sample size, indicating excellent randomization. The treatment group had a average age that was about 10.5 years old than the average age of the control group. However, since the lowest age was 20 and the highest age was 80, so a difference of only 10.5 years–although slightly high–still does reflects a success in generalization as when we break age down by group visually in the boxplot below, we see the ranges overlap with the median of the treatment being within the third quartile of the control.



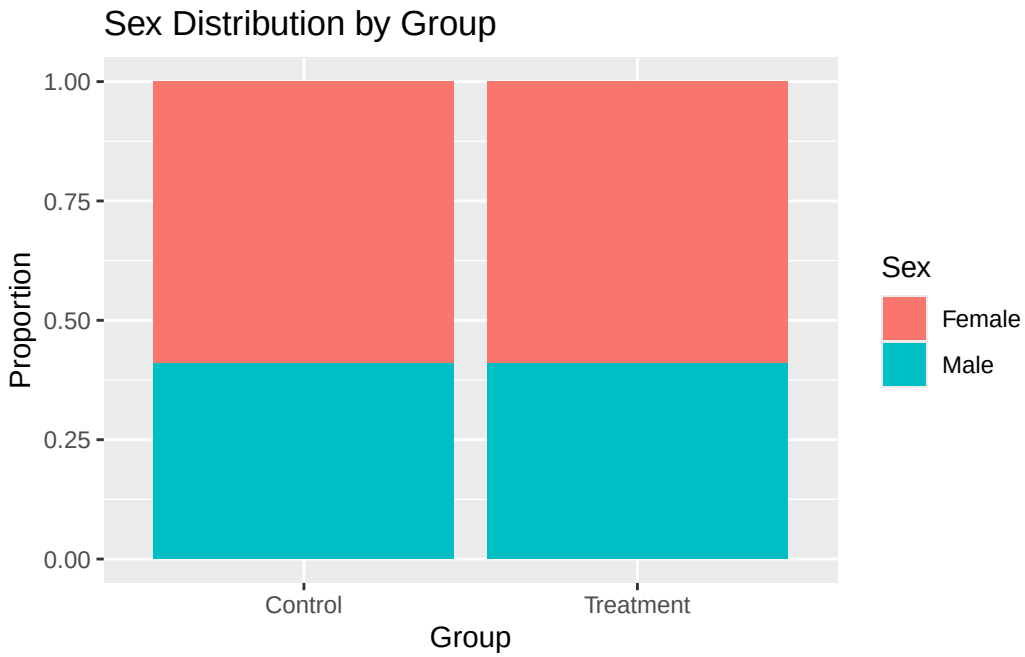
We saw a difference of 370.41 dollars in individual wealth between the two groups. Although this is numerically larger than other comparisons, we expect this sort of variation for a variable like wealth, which ranged from 265 dollars to 13,171 dollars. Overall, our power analysis confirmed that our randomization worked, thus allowing for generalization in the conclusions of this study.

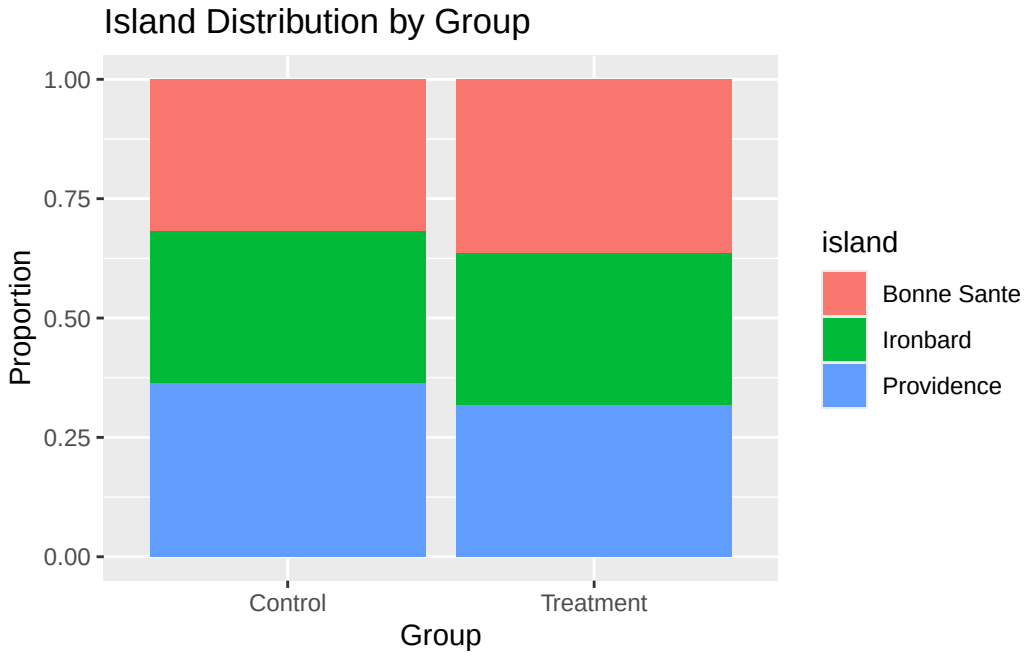
Visual and Numerical Summaries of the Variables Collected

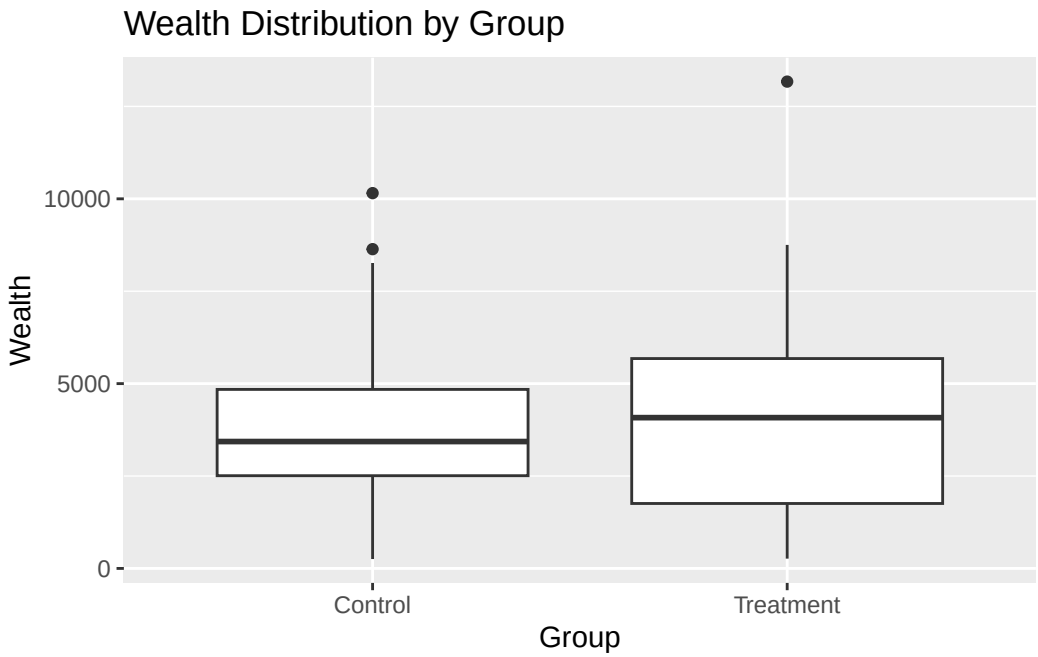
Below is a numerical summary table indicating differences in the variables collected between the explanatory groups.

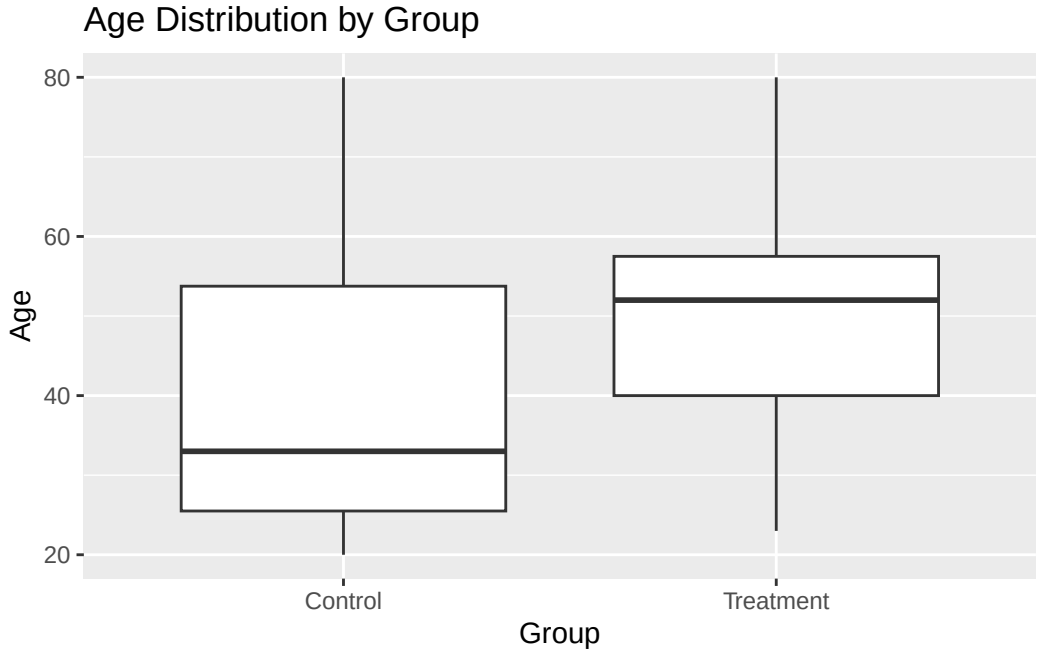
```
                                                              Measure     Value
1 Difference in Sex Ratio (male/female) between Treatment & Control   0.00000
2           Difference in Median Wealth between Treatment & Control 370.40910
3                Difference in Mean Age between Treatment & Control  10.54545
```

We have also provided figures to better visually understand the distribution of the variables collected.


Sex Distribution by Group
1.00
0.75
0.50
0.25
0.00
Proportion
Control
Treatment
Group
Sex
Female
Male


Island Distribution by Group
1.00
0.75
0.50
0.25
0.00
Proportion
Control
Treatment
Group
island
Bonne Sante
Ironbard
Providence





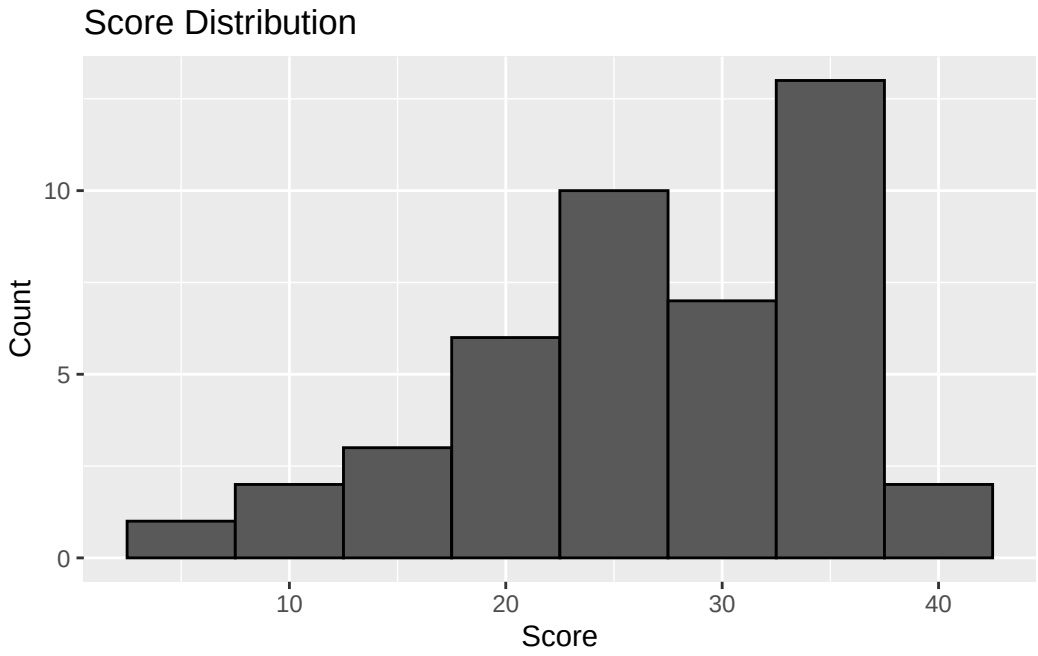
Hypothesis Test

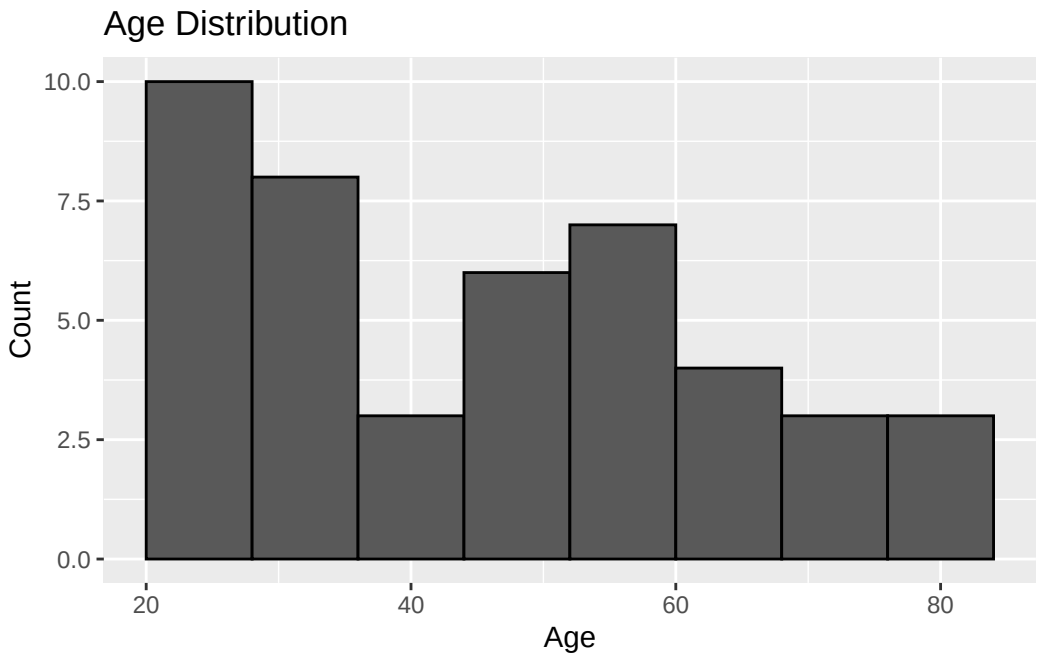
Hypothesis Test to check for Randomization

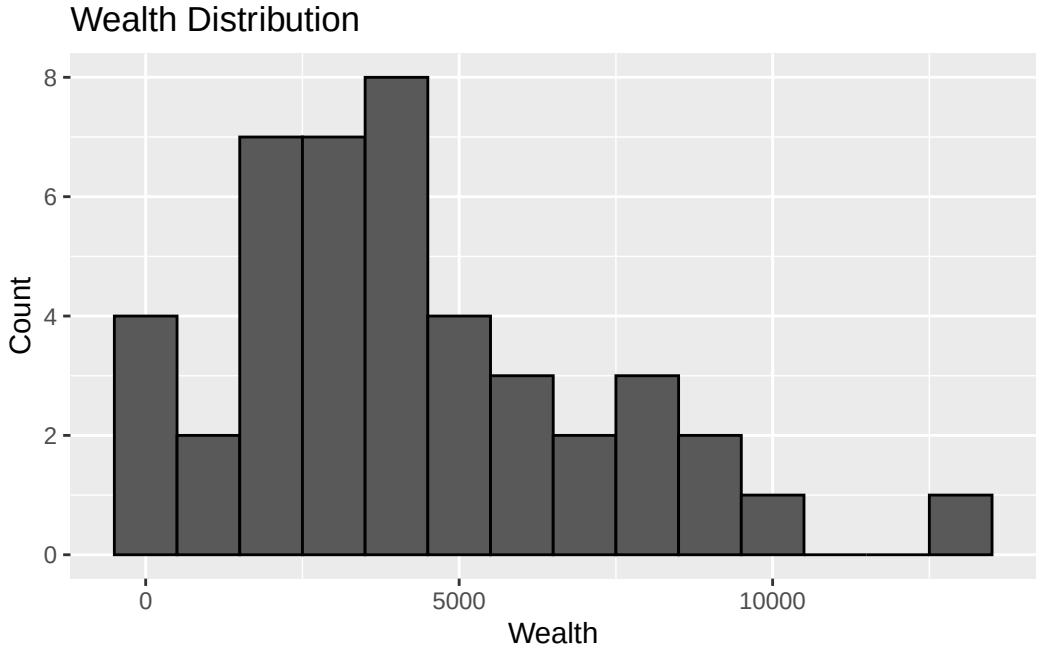
Our Null Hypothesis for each variable: No true difference in that variable across our control and experimental group or no true relationship between that variable and the two groups (independent relationship).

Alternative Hypothesis: There is a difference in that variable across our control and experimental group or there is a relationship between that variable and the two groups.

Check for Conditions: To check the variables meet the necessary conditions for these tests we have provided histograms of the continous variables below to show the distributions.







From the distributions above, we see that the distribution of scores, ages, and wealth does have some skewness; however, we should still be able to proceed with our t-test as we have a large sample size (45), and the CLT applies, therefore, we will not be violating the normality condition. Otherwise, we have satisfied random sampling and independence of observations

with the way our data has been collected.

For the variable of sex, we meet the conditions for the Chi-square test for sex, as we have a random sample, at least 1 observation in every cell, and at least 5 expected counts in each cell. Consider the Chi-Squared table below to confirm this.

We can now conduct t-tests for age and wealth across the treatment and control group and also conduct a chi-squared test for sex. Below is the output of our tests.

```
    Welch Two Sample t-test

data:  age by mushrooms
t = -2.0121, df = 41.809, p-value = 0.05068
alternative hypothesis: true difference in means between group NO and group YES is not equal
95 percent confidence interval:
 -21.12350565   0.03259656
sample estimates:
 mean in group NO mean in group YES
         40.59091          51.13636


    Welch Two Sample t-test

data:  wealth by mushrooms
t = -0.41348, df = 39.902, p-value = 0.6815
alternative hypothesis: true difference in means between group NO and group YES is not equal
95 percent confidence interval:
 -2181.109  1440.291
sample estimates:
 mean in group NO mean in group YES
         4023.318          4393.727


    Pearson's Chi-squared test

data:  table(select(final_data, sex, mushrooms))
X-squared = 0, df = 1, p-value = 1

        mushrooms
sex      NO YES
  Female 13  13
  Male    9   9
```

From the following above, we observe p-values greater than 0.05 for all of our variables compared across our experimental and control groups, suggesting that we have evidence that fails to reject the null from above. This confirms that our randomization worked and there are no discernible differences in the variables across treatments.

Hypothesis Test for Intial results of Research Question

Our Null Hypothesis: There is no difference in the true average arithmetic scores between those who consumed Psilocybin mushrooms and those that did not.

Alternative Hypothesis: There is a difference in the true average arithmetic scores between those who consumed Psilocybin mushrooms and those that did not.

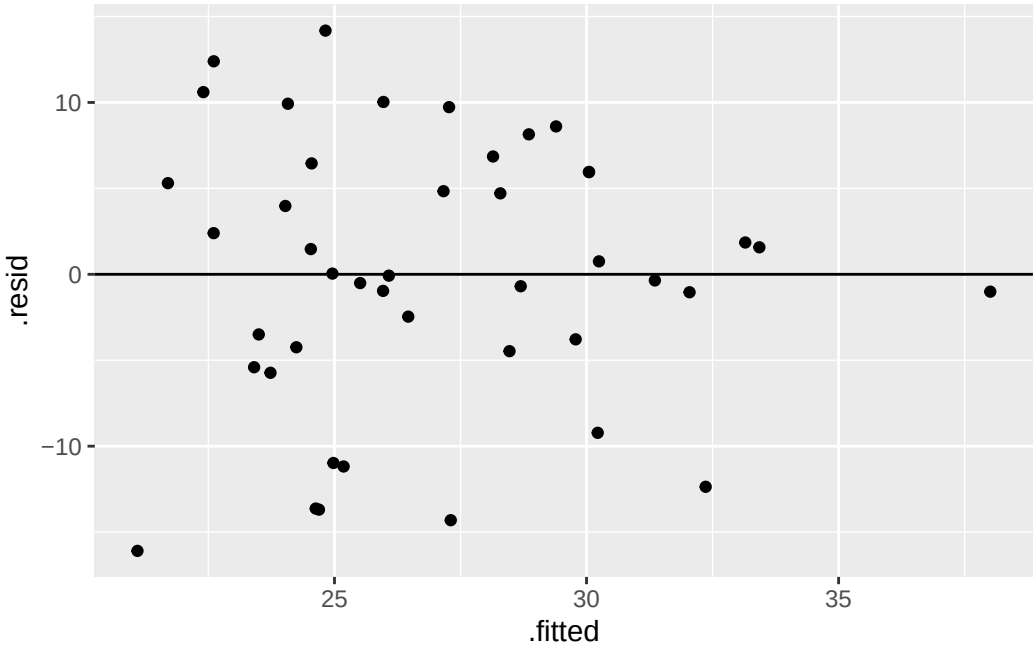
```
    Welch Two Sample t-test

data:  score by mushrooms
t = -1.529, df = 41.682, p-value = 0.1338
alternative hypothesis: true difference in means between group NO and group YES is not equal
95 percent confidence interval:
 -9.069764  1.251582
sample estimates:
 mean in group NO mean in group YES
         25.00000          28.90909
```

From the following, we observe a p-value greater than 0.05–0.1338 $>$ 0.05–indicating that we fail to reject the null hypothesis and we cannot conclude that there is a discernible difference in true average arithmetic scores between those who consumed psilocybin mushrooms and those that did not.

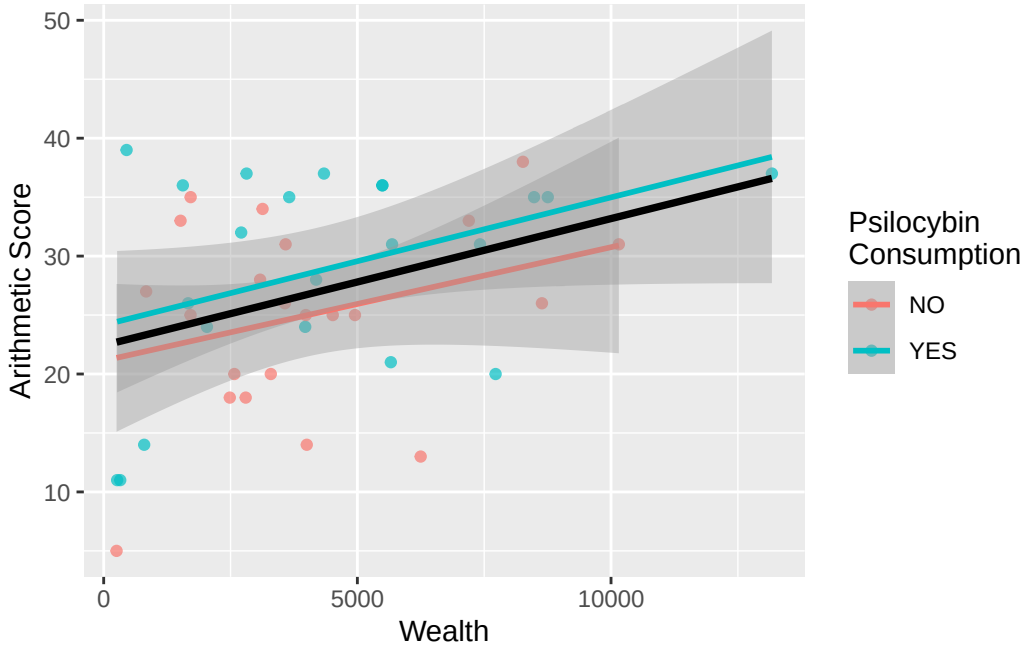
Multiple Linear Regression Model

```
# A tibble: 3 x 5
  term         estimate std.error statistic  p.value
  <chr>           <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)  20.8      2.39          8.73 6.83e-11
2 wealth        0.00104  0.000415      2.50 1.66e- 2
3 mushroomsYES  3.53     2.42          1.46 1.52e- 1
```



The goal of the multiple linear regression model is to determine whether psilocybin mushroom consumption affects exam scores, so we have included mushrooms as a variable because it is the explanatory variable that our research question focuses on. Wealth is the other variable that is included because it has a p-value below 0.05 so wealth is statistically discernible. Other variables that were collected were excluded from the model as they had high p-values suggesting that they were not statistically discernible. Now from this model, suppose we run into an islander who has a wealth of 500 and who consumed the psilocybin mushroom before the test. We can obtain a 95% prediction interval of (8.04, 41.70) which is centered at score $= 20.83 + 0.00104(500) + 3.53 = 24.88$. So This means that we would expect the arithmetic score for an individual with these characteristics to fall between 8.04 and 41.70 about 95% of the time. It is important to note that 40 is the highest possible score an individual can obtain for a difficult arithmetic test in The Islands so although the prediction interval calculation results in a greater upper bound, this doesn't necessarily mean an individual score can be that high. The variable of wealth is statistically discernible since the p-value is below 0.05 so it is a discernible predictor of score and the coefficient associated with wealth means that per dollar of health, given that all else is constant, the score increases by 0.00104. The variable of whether the individual consumed psilocybin mushrooms had a p-value of 0.15 which is far greater than 0.05 suggesting the variable is not statistically discernible and we can not conclude being in the treatment or control group causes higher scores. After controlling for wealth, there is no statistically significant evidence that consuming 10 grams of psilocybin mushrooms affects mental arithmetic exam scores among adults on the Islands. However, wealth is a significant predictor of performance as shown by the p-value in the model we have. To visualize the significance of wealth in predicting score, consider the model below that depicts

wealth along with score with treatment group represented through color. There are three prediction regression lines below. The black line represents the model for all participants wealth in relation to predicting score. The other lines represent the model if accounting for mushroom consumption to understand the effect of considering mushroom consumption in the final model. The visual model below shows that the shaded confidence bands for each treatment group overlap significantly and the colored regression line seems to fall within the confidence band of the other regression line. This indicates that the estimated effect of mushroom consumption is small relative to its standard error, which explains why the p-value for mushrooms is relatively high.



Limitations

As mentioned in the methods section, we attempted to conduct random sampling to the best of our ability. We used a random number generator to select a city and a household on an island, and then randomly pick an adult from that household to be a participant. Although we attempted to randomly sample all adults in The Islands, it was not truly random. Since we tried to sample roughly the same number of individuals from each island, our sample may not perfectly represent the actual population distribution of The Islands. For example, if one island has a much larger adult population than another, then sampling the same number of people from both islands could over represent adults from smaller islands and under represent adults from larger islands. Therefore, our results may not generalize perfectly to all adults on The Islands.

As seen from the randomization hypothesis tests above, the variables we collected seem to be roughly balanced across the two groups. In other words, our random assignment process was effective in distributing the variables across the treatment and control group, and in reducing the concern of confounding variables. With this knowledge, we can state that we do not expect major limitations in our ability to infer a causal relationship between psilocybin mushroom and arithmetic scores, due to the effectiveness of the random assignment process. However, this must be taken with caution as this experiment calculated a non-significant p-value, that does not support a casual relationship between the two variables.

One potential limitations of this study that we believe might hinder our ability to answer the research claims we set out to study is the sample size. Although we calculated a non-significant p-value, a closer inspection of the data revels that the mean arithmetic score for those in the treatment group is higher than that of the control group. While it is possible that this difference can simply be due to the natural variability of the data–the argument for the non-significant p-value–it could also suggest an underlying trend that becomes significant at a larger sample size. Thus, taking a larger sample might provide a better answer to the relationship between psilocybin mushroom and arithmetic scores, making the sample size a potential limitation of this experiment.

Results & Conclusions

Ultimately, since our p-value was not discernible at a 95% discernibility level, we failed to reject the null hypothesis. That is, we failed to reject the conclusion that there is no true difference in average test score on a difficult arithmetic test between individuals aged 18+ on the Islands who consume psylocybin mushrooms and those who do not. This does not, however, mean that we conclude the null hypothesis is true, rather that we are unable to make any concrete claims based on our data. Note that, although we are not able to make any conclusions, our study was crafted so that, had we been able to reject the null, we would have been able to generalizatize to the population due to our randomization.

Despite our inability to make a claim regarding our research question, we encourage future replication of this study but with some added additions that might better provide information about the relationship between the two variables. This may manifest as collecting information on level of education of participants or using different dosages of mushroom–as various dosages have been known to have various effect on cognition (Bonnieux et al.)[2]–or even conducting a paired t-test to reduce the between-subject variability. Although there is always room for improvements and further additions, and despite our lack of a conclusive claim, we hope that the analysis conducted as part of the process is able to provide a future framework for analyzing this, or related research questions.

2

[2]Bonnieux et al.